INCH POUND
MIL-PRF-29610
25 January 1996

PERFORMANCE SPECIFICATION

VISOR ASSEMBLY, MULTIPLE WAVELENGTH LASER PROTECTIVE, AVIATORS

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers the requirements for aircrew eye protective visor assemblies which provide protection against multiple fixed laser wavelengths. The visors may be used by all fixed and rotary wing aircrew.

1.2 Classification. The type designators of the visor assemblies covered by this specification are as follows:

EEU-8/P (HGU-68/P fixed wing aircrew helmet assembly)
EEU-12/P(HGU-12P and HGU-67/P rotary wing helmet assembly)
EEU-13/P(HGU-68/P(V) fixed wing helmet assembly)
EEU-14/P(HGU-68/P fixed wing aircrew helmet assembly)
EEU-15/P(HGU-87/P fixed wing aircrew helmet assembly)
AIHS (HGU-56/P rotary wing helmet assembly)
EOHSS (HGU-56/P rotary wing helmet assembly)

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in Sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed as applicable documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Systems Standardization, Code 4.1.4B120-3 Naval Air Warfare Center Aircraft Division, Highway 547, Lakehurst, NJ 08733-5100, by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D1003 - Standard Test Methods for Haze and Luminous Transmittance of Transparent Plastics

ASTM-D1044 - Standard Test Method for Resistance of Transparent Plastics to Surface Abrasion

ASTM-D3935 - Standard Specification for Polycarbonate (PC) Unfilled and Reinforced Material

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1137)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in the document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Individual visor requirements. The individual visor requirements shall be as specified herein.

3.2 First article inspection. When specified, (6.2) samples shall be subjected to first article inspection (6.4) in accordance with 4.3.

3.3 Materials. Materials shall conform to the applicable procurement documents as specified herein. Materials which are not covered by these documents or are not specifically described herein, shall be of the best quality, of the lightest practical weight and suitable for the purpose intended. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Visor material. Unless otherwise specified (see 6.2), the polycarbonate material used for molding the visor substrate shall be in accordance with ASTM D3935-87.

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3.4 Design and dimensions. The design and dimensions of the visors shall be as specified herein. The visor shall have the same physical dimensions, fir and function as the visors identified in figures 1 through 7. The visor shall afford protection for aircrew against the specified wavelengths to the required densities (6.2) throughout the entire safety zone (see 4.5.4.2). The front and back surfaces of the visor shall have a clear coating which meets the abrasion resistance, adhesion and environmental requirements specified herein. The front and back surfaces of the visor shall have an anti-reflection coating. All cockpit displays, phosphors (P43) and lights shall be visible when the visor is worn.

3.4.1 Area of vision. Unless otherwise specified (see 6.2), the visor shall have three areas of vision which are identified as: critical area, non-critical area and the area outside of these areas of vision (no optical requirement). The center point for the left and right critical areas (area 1L & 1R) shall be identified as "design eyepoint". Figure 8 shows a typical layout of a visor and is intended to serve as a guide for purposes of determining matching areas of the left and right sides of the visor. The left and right design eyepoints as designated in Figure 8, when tested as specified in 4.5.3 shall match in prism, power and transmittance to the requirements of 3.7.1 and 3.7.2.

3.5 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.6 Support equipment. If an eye centered design is used, an overlay template is required. The template shall locate the left and right design eyepoints of the visor (4.5.7) and allow these locations to be temporarily marked in order to support fitting and adjustment of the visor.

3.7 Performance.

3.7.1 Optical.

3.7.1.1 Prismatic deviation. Prismatic deviation stated herein includes the inherent prismatic power resulting from non-parallel surfaces of the material and dissimilar curvature between the critical areas. The vertical and horizontal prismatic deviations shall meet the requirements of 3.7.1.1.1 and 3.7.1.1.2 when tested as specified in 4.5.3.1.

3.7.1.1.1 Vertical prismatic deviation. Prismatic power when measured along the line of sight (i.e. in the worn position) at the design eyepoint for the right eye and the design eyepoint for the left eye as specified in 4.5.3.1 shall not:

a. Exceed a prism imbalance of .125 diopters for prism of the same or opposite base (i.e. base up - base up $\leq$ .125, or base down - base down $\leq$ .125 or base up - base down $\leq$ .125)

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b. Exceed .25 diopters of base up/base down for either critical area, or exceed a total of .5 prism diopters of the same base (i.e., base up + base up ≤ .5).

3.7.1.1.2 Horizontal prismatic deviation. Prismatic power, when measured along the line of sight (i.e., as in the worn position) at the design eyepoint for the right and left eye as specified in 4.5.3.2, shall not:

a. Exceed a prism imbalance of .125 diopters base in (base in - base in ≤ .125) or .25 diopters base out (base out - base out ≤ .25).

b. Exceed .25 diopters for base in/ for either critical area, or exceed a total of .5 prism diopters of the same base (baseout + base out ≤ .5)

3.7.1.1.3 Spherical & cylindrical power. The spherical and cylindrical power when measured as specified in 4.5.3.3 shall not exceed plus or minus 0.125 diopters or differ in sign at left and right optical center (i.e. positive power at one optical center and negative power at the other optical center).

3.7.1.1.4 Optical distortion in the critical areas. The visor shall be free from irregularities affecting optical qualities which would cause sudden jumps and blurring of the visual scene when viewed through the visor. The distortion within the critical vision area when examined as specified in 4.5.3.4 shall be within the acceptable limits shown in Figure 9. This is a pass/fail determination.

3.7.2 Transmittance.

3.7.2.1 Photopic and scotopic transmittance. When tested as specified in 4.5.4.1, the photopic transmittance throughout the critical area shall be 30% or less. The scotopic transmittance shall be 20 % or greater. Over the surface of the visor, these transmittances (i.e. left photopic vs. right photopic and left scotopic vs. right scotopic) shall not differ by more than 10% of the higher of the two values. The transmittance at the top of the visor shall not exceed the transmittance at the bottom of the visor.

EXAMPLE OF PASSING SAMPLE:	Right eye photopic	=	18.95%
	Left eye photopic	=	17.5 %
	18.95 x .10	=	1.895%
	18.95 - 18.50	=	1.45% (within the allowable difference of 1.895%)
EXAMPLE OF FAILING SAMPLE:	Right eye photopic	=	18.75%
	Left eye photopic	=	16.5%
	18.75 x .10	=	1.875%
	18.75 - 16.50	=	2.25% (outside the allowable difference of 1.875%)

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3.7.2.2 Optical density and angular protection. The visor shall protect the human eye against specified multiple fixed laser wavelengths when tested as specified in 4.5.4.2 The visor shall have the required optical density at the specified threat wavelengths (see 6.2) within the safety zone (see 4.5.4.2).

3.7.2.3 Erythemal ultraviolet transmittance. The erythemal ultraviolet transmittance of the visor, when computed as specified in 4.5.4.3 shall not exceed 0.3%.

3.7.2.4 Haze. The visor haze shall not exceed 3.0% when determined as specified in 4.5.4.4.

3.7.3 Environmental.

3.7.3.1 Radiation induced damage resistance. When tested as specified in 4.5.5.2.1 and 4.5.5.2.2, transitory or permanent radiation induced changes in the density or transmittance of the visor shall not occur at levels typical of those which aircrew may be exposed during operation.

3.7.3.2 High temperature. The visor shall withstand high temperature stresses when tested as specified in accordance with 4.5.5.3.

3.7.3.3 Low temperature. The visor shall withstand low temperature stresses when tested in accordance with 4.5.5.4.

3.7.3.4 Temperature shock. The visor shall withstand the effects of rapid changes in ambient temperature when tested in accordance with 4.5.5.5.

3.7.3.5 Rain. The visor shall withstand the effects of prolonged exposure to rain when tested in accordance with 4.5.5.6.

3.7.3.6 Humidity. The visor shall be capable of being stored or used under wide temperature and humidity extremes when tested in accordance with 4.5.5.7.

3.7.3.7 Salt fog . The visor shall withstand the effects of prolonged salt fog when tested as specified in 4.5.5.8.

3.7.3.8 Chemical. When tested as specified in 4.5.5.9, the visor shall withstand exposure to the following chemicals:

a. Insect repellent, controlled release (DEET)
b. JP4 aircraft turbine fuel
c. JP8 aircraft turbine fuel, kerosene type
d. Aircraft lube oil, turbine engine, synthetic base
e. Aircraft hydraulic fluid, petroleum base
f. Aircraft hydraulic fluid, fire-resistant, synthetic base
g. Degreasing solvent, type I (mineral spirits)
h. Cleaning compound, turbine engine gas path, 20% solution

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3.7.4 Mechanical.

3.7.4.1 Ballistic resistance. After being tested as specified in 4.5.6.1, the witness sheet following three valid impacts on the visor shall be examined for any impact or penetration. Impact or penetration of the witness sheet shall denote failure.

3.7.4.2 Abrasion resistance. After being subjected to the abrasion test specified in 4.5.6.2 the haze shall not exceed 6% and the transmittance shall not decrease by more than 4 % (for example, an initial value of 20% transmittance cannot decrease to less than 16%).

3.7.4.3 Adhesion of coating. The coating, when tested as specified in 4.5.6.3 shall not be removed, dislodged, or affected in any way. Removal or loosening of the coating shall be cause for rejection.

3.7.4.4 Windblast resistance. The EEU-8/P, EEU-13/P, EEU-14/P and EEU-15/P visor assemblies (visor with guides attached), when tested as specified in 4.5.6.4 shall remain in the full down and locked position and there shall be no visible evidence of material failure on the rear surface.

3.8 Weight. The visor shall weigh as little as possible without compromising the protection levels.

3.9 Identification of product. Each visor shall be identified by type designator, drawing/part number, lot number and additional identification that shall allow tracking of each visor from the time of final assembly until disposal. The identification shall not cause the visor to be classified. The identification marking shall not interfere with the optical or protective properties of the visors. If an eye centered approach is used and more than one visor size is required, the visor size shall also be permanently marked on the visor. All letters and numbers shall be white, bold and vertical on clear coated black pressure sensitive high adhesive vinyl film. The label dimensions and location of the required information are shown in figure 10.

3.10 Workmanship. The visor shall conform to the quality of product established by this specification. All visors shall be constructed to produce an item free from all defects which would affect proper functioning in service when tested as specified in 4.5.8 and shall be manufactured in accordance with the following criteria:

3.10.1 Surface defects. The visor shall meet the requirements of paragraphs 3.10.1.1 through 3.10.1.4 when tested as specified in 4.5.8.

3.12.1.1 Definition of surface defects.

Blemish	Any pit, scratch, dig, bubble, inclusions, etc. having a diameter greater than 0.015" but less than 0.035" or a length up to 0.100" and a width up to 0.010".

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Flaw — Any pit, scratch, dig, bubble, inclusion, etc. having a diameter greater than 0.035" but less than 0.050" or a length up to 0.150".

Gross flaw — Any pit, scratch, dig, bubble, inclusion, etc. having a diameter greater than 0.050" or a length greater than 0.150".

Cluster — A grouping of three or more blemishes within a 0.5" diameter circle.

3.10.1.2 Critical viewing area. There shall be no more than five (5) blemishes in the critical viewing area. There shall be no flaws, gross flaws or clusters in the critical viewing area.

3.10.1.3 Non-critical viewing area. There shall be no more than 10 blemishes, 5 flaws or 3 clusters (not to exceed a total of 10 blemishes) or some combination not to exceed a total affected area of 0.35 inch in the non-critical viewing area. There shall be no gross flaws in the non-critical viewing area.

3.10.1.4 Area outside the area of vision. There are no requirements for the area outside the area of vision.

3.10.2 Other defects. The visor shall meet the requirements of 3.10.2.1 through 3.10.2.7 when tested as specified in 4.5.8.2.

3.10.2.1 Cloudiness in the critical area of vision. The visor shall have no cloudiness in the critical area of vision, where cloudiness is defined as any included opacity, turbidity, darkness, obscurity, dimness or milkiness. If cloudiness is observed, the visor shall be tested for compliance with 3.7.2.4.

3.10.2.2 Crack, hole, chip or break in the visor material. The visor shall have no cracks, holes, chips or breaks in the visor material in any area of the visor.

3.10.2.3 Warping. The visor shall have no evidence of warping, buckling or twisting. If warping is apparent, the visor shall be tested for the dimensions and compatibility requirements of 3.4.

3.10.2.4 Striations or waviness in the critical area of vision. The visor shall have no striations or waviness in the critical area of vision, where striations are defined as two or more lines, streaks, stripes, bars, bands, or other lineation, generally (but not necessarily) parallel. If striations are observed in the visual examination the visor shall be tested for compliance with 3.7.1.1.4.

3.10.2.5 Surface abrasion, dullness or roughness in the critical area of vision. The visor shall have no surface abrasions or areas of dullness or roughness in the critical area of vision, as evidenced by any surface area that appears to be worn due to friction, rubbing, wear, chafing, scraping, grazing, scratching, scuffing, scrubbing or scouring.

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3.10.2.6 Stain or discoloration. The visor shall have no spot, blotch, patch, mottle or dapple that appears to be the residue of some material that has evaporated or otherwise been deposited on the surface or any discoloration not readily removed by water in the critical area of vision. "Not readily removed by water" is defined as not removable using a soft cloth saturated with distilled water and gently rubbing the area for at least 5 seconds but not more than 15 seconds.

3.10.2.7 Sharp edge or burr on an edge. The visor shall have no sharp edges or burrs.

4. VERIFICATION.

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Conformance inspection (see 4.4)

4.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the applicable test method or test paragraph in this specification.

4.3 First article inspection. First article inspection shall consist of all the examinations and tests specified in Table I. Failure of any sample to be in accordance with the applicable figures and requirements for which it is being tested will result in failure of the lot from which the sample is drawn.

4.3.1 First article samples. Unless otherwise specified the manufacturer shall submit samples of complete, assembled visors and, if applicable, support equipment. The samples shall be representative of the construction, workmanship, components, and materials to be used during production. The production qualification inspection samples shall be furnished to the Government as directed by the contracting officer (6.2) and shall not be counted as part of the quantity to be delivered as part of the contract. The visor samples will be consumed or destroyed in the first article testing

4.3.1.1 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made during the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.4 Conformance inspection. Conformance inspection shall consist of the examinations and tests specified in Table III. The number of samples to be inspected per lot size and inspection level shall be determined from Table II. Failure of a sample to be in accordance with the applicable figures and requirements of Table II shall result in rejection of the sample.

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4.4.1 Inspection lot formation. The inspection lot shall be defined as all of the same type (part number) visor assemblies produced under the same conditions using the same materials and components as the production qualification units in accordance with the same drawings and specifications. The inspection lot shall be identified by the contractor and shall be kept intact as a lot for inspection purposes.

4.5 Test results. Test results shall include the following information:

a. Contractor identification
b. Test facility
c. Contract number
d. Lot numbers and quantities
e. Visor specification number
f. Date
g. Personnel conducting test and any witnesses
h. Remarks pertinent to the conduct of the test, or behavior of the material or item

4.5.1 Component and material certification. A Certificate of Conformance (COC) shall be acceptable as evidence that the materials conform to the requirements of 3.3 and 3.3.1.

4.5.2 Design and Dimensions. The visor shall be examined for conformance to the design and dimensions specified in 3.4 and 3.4.1. Any failure to meet these requirements shall be classified as a critical defect.

4.5.3 Optical inspections. Prior to performing any of the following tests, the visor shall be visually checked for defects such as cracks, chips and delamination. Before and after subjecting the visor to any tests, it shall be thoroughly and carefully cleaned to remove dirt, oil, fingermarks, etc., to assure that all the tests are performed on a clean visor. Matched locations (Figure 8) shall be tested for sign, prism and power with a suitable instrument capable of resolving increments of 0.06+ diopter. The visor shall pass the matching requirements specified in 3.4.1.

4.5.3.1 Vertical prismatic deviation. The visor shall be tested for vertical prismatic deviation as described in 4.5.3. Base up prism shall be designated positive (+) and base down prism shall be designated negative (-). The vertical prismatic deviation shall be calculated by determining the algebraic sum and difference between the amount of prism at design eyepoint of the right eye and design eyepoint of the left eye and shall pass the requirements specified in 3.7.1.1.1.

4.5.3.2 Horizontal prismatic deviation. The visor shall be tested for horizontal prismatic deviation as described in 4.5.3. Base out prismatic deviation shall be designated positive (+) and base in prismatic deviation shall be designated negative (-). The horizontal prismatic deviation shall be

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calculated by determining the algebraic sum and difference between the amount of prism at design eyepoint of the right eye and the amount of prism at design eyepoint of the left eye and shall pass the requirements specified in 3.7.1.1.2.

4.5.3.3 Spherical and cylindrical power. The visor shall be tested for spherical and cylindrical power using the method described in 4.5.3 and shall pass the requirements specified in 3.7.1.1.3.

4.5.3.4 Optical distortion. With the visor inserted into a suitable apparatus (a schematic of a sample apparatus is shown in Figure 11) with its surface normal to the line of sight, examine for distortion. The degree of off-parallelism will constitute the amount of distortion and this distortion pattern shall be compared with Figure 9. The visor shall meet the requirements specified in 3.7.1.1.4.

4.5.4 Transmittance. The spectral transmittance shall be measured using a spectrophotometer from 270 to 1200 nm in increments of 1 nm or less. The photopic and scotopic transmittance of the visors shall be computed as shown in Table IV or using an equivalent method. The calculations may be done concurrently or with the use of a computer or programmable calculator.

4.5.4.1 Photopic and scotopic transmittance. When tested as specified in 4.5.4, the visor shall pass the requirements specified in 3.7.2.1.

4.5.4.2 Optical density. The optical densities shall be measured at all angles at which incident laser radiation in the specified threat wavelength bands may enter the eye. The calculation for angular protection shall be for an eye with an 8 mm diameter pupil rotated ± 35 degrees with respect to the optical axis of the lens at the appropriate eye relief. The measurements shall be made with a spectrophotometer, or a photometer and laser filtered light sources of the appropriate wavelength(s). The densities shall pass the requirements specified in 3.7.2.2.

4.5.4.3 Erythemal ultraviolet transmittance. The erythemal ultraviolet transmittance of the visor shall be computed by averaging the transmittance at 290, 300, 310 and 320 nm. The visor shall pass the requirements specified in 3.7.2.3.

4.5.4.4 Haze. The haze shall be determined in accordance with Procedure A of ASTM D1003. The visor shall pass the requirements specified in 3.7.2.4.

4.5.5 Environmental tests.

4.5.5.1 General test performance guidance. Before performing the environmental tests, all the test visors shall be measured and meet the requirements for transmittance (3.7.2.1), optical density (3.7.2.2), and lens defects (3.10). Visors shall be placed in the environmental chamber to allow adequate circulation around the item. Interruptions during which the prescribed test tolerances are not exceeded shall be considered as part of the total test

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duration. Videotape records may be substituted wherever photographs are required in the test procedures.

4.5.5.2 Radiation induced damage resistance.

4.5.5.2.1 Solar radiation. The visor shall be placed in a solar simulator. The total incident flux shall be monitored and recorded throughout the exposure. The visor shall be tested as outlined below:

Step 1 - Prepare the visor in accordance with general test performance guidance 4.5.5.1.
Step 2 - Adjust the solar radiation source to a radiant energy rate of 1120 ± 47 W/m^2 (355 ± 14 Btu/ft^2/hr) and a temperature of 49°C (120°F).
  a. Photograph the visor after temperature stabilization.
  b. Photograph the visor at 2 hour intervals for the remainder of the 24 hour period.
Step 3 - Maintain these conditions for 20 hours
Step 4 - Measure and record the test item temperature.
Step 5 - Turn off the solar radiation source for 4 hours.
Step 6 - Measure haze and transmittance.
Step 7 - Repeat steps 2 through 5 for 11 cycles

At test completion, the visor shall pass transmittance (3.7.2.1 and 3.7.2.3), optical density (3.7.2.2), abrasion resistance (3.7.4.2, Adhesion coating (3.7.4.3), defects (3.10) and Ballistic resistance (3.7.4.1).

4.5.5.2.2 Laser saturation. The visor shall be scanned with a spectrophotometer before and after the laser saturation tests. The visor shall be exposed to laser energies at the protective wavelengths starting at an intensity below which any bleaching and/or damage will occur. The intensity shall be increased until reversible saturation (temporary bleaching) and/or damage to the visor occurs. The results shall be stated in terms of wavelength and energy density at which bleaching and/or damage occur.

4.5.5.3 High temperature. The visor shall be tested in accordance with paragraph 4.5.5.1 and as outlined below:

Step 1 - Prepare the visor in accordance with paragraph 4.5.5.1.
Step 2 - Place the visor in the test chamber ensuring it is positioned so the entire front surface can be photographed from the viewing port without disturbing the sample. Place a white light behind the sample so the illumination is uniform throughout the sample.
Step 3 - Raise the internal temperature to 100°F (38°C).
  a. After temperature stabilization, photograph the visor.
Step 4 - Maintain internal temperature at 100° ± 2°F (38° ± 1°C) for 6 hours.

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b. Photograph the visor at the end of the 2nd and the end of the 6th hours

Step 5 - Raise temperature to 151°F (66°C) over a 1 hour time period and maintain that temperature for an additional 4 hours.
a. Photograph visor after stabilizing the temperature at 151°F.
b. Photograph visor at the end of the 2nd and 4th hours.

Step 6 - Lower chamber temperature to 100° ± 2 °F (38° ± 1°C) within a time period of 1 hour. a. Photograph the visor after temperature stabilization.

Step 7 - Repeat steps 4, 5 and 6 two additional times.

The visor shall pass the requirements of 3.7.3.2. Following the high temperature test, the visor shall pass the requirements of 3.7.1 and 3.7.2.

4.5.5.4 *Low temperature*. The visor shall be tested in accordance with paragraph 4.5.5.1 and as outlined below:

Step 1 - Prepare the visor in accordance with paragraph 4.5.5.1.

Step 2 - Place the visor in the test chamber ensuring it is positioned so the entire front surface can be photographed from the viewing port without disturbing the sample. Place a white light behind the visor so the illumination is uniform throughout the sample.

Step 3 - Lower the internal chamber temperature to the storage temperature of -60° ± 2°F (-51° ± 1°C) and maintain for a period of 24 hours after stabilization.
a. Photograph the visor after temperature stabilization.
b. Photograph visor at 2 hour intervals for the remainder of the 24 hour period.

Step 4 - Return visor to standard ambient conditions and stabilize.
a. Photograph visor after temperature stabilization.

The visor shall pass the requirements of 3.7.3.3. Following the low temperature test, the visor shall pass the requirements of 3.7.1 and 3.7.2.

4.5.5.5 Temperature shock. The visors shall be tested in accordance with 4.5.5.1 and the procedure outlined below:

Step 1 - Prepare the visor in accordance with paragraph 4.5.5.1.

Step 2 - Place the visor in the test chamber ensuring it is positioned so the entire front surface can be photographed from the viewing port without disturbing the sample. Place a white light behind the visor so the illumination is uniform throughout the sample.

Step 3 - Raise the internal chamber temperature to 151° ± 2°F (66° ± 1°). Maintain for a period of 4 hours.
a. Photograph the visor after temperature stabilization.

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b. Photograph the visor at the end of 2 hours and at the end of 4 hours.

Step 4 - Transfer the visor within 5 minutes to a cold chamber with an internal temperature of -60° ± 2°F (-51° ± 1°C) . Expose the visor to this temperature for a period of 4 hours.
a. Photograph the visor within 5 min. of placement in cold chamber.
b. Photograph the visor at the end of 2 hours and at the end of 4 hours.

Step 5 - Return the visor within 5 minutes to the high temperature chamber. Expose the visor to this temperature for a period of 4 hours.
a. Photograph the visor as soon as it defrosts.
b. Photograph the visor at the end of 2 hours and at the end of four hours.

Step 6 - Repeat step 4.
Step 7 - Repeat step 5.
Step 8 - Return visor to standard ambient and stabilize.
a. Photograph visor after temperature stabilization.

The visor shall pass the requirements of 3.7.3.4. Following the temperature shock test, the visor shall pass the requirements of 3.7.1 and 3.7.2.

4.5.5.6 Rain. The visor shall be tested in accordance with paragraph 4.5.5.1 and the procedure outlined below. The visor configuration is "unprotected". This choice of test is based on the fact that the visor may be exposed but is normally protected from rain.

Step 1 - Prepare the visor in accordance with paragraph 4.5.5.1.
Step 2 - Place the visor in the test chamber ensuring it is positioned so the entire front surface can be photographed from the viewing port without disturbing the visor. Place a white light behind the visor so the illumination is uniform throughout the visor.
a. Photograph the visor.
Step 3 - The rainfall rate shall be 1.89 in/hr (0.80 mm/min).
Step 4 - The droplet size shall be between 0.5mm and 4.5mm.
Step 5 - The water temperature shall be 77° ± 18°F (25° ± 10°C).
Step 6 - The visor shall be preheated a minimum of 17°F (10°C) higher than the water temperature.
Step 7 - The chamber temperature shall be uncontrolled except as regulated by the water temperature.
Step 8 - Expose the entire outer surface of the visor the rain.
Step 9 - Expose the visor for two hours of rain.
a. Photograph the visor at the half-way point of the exposure time.
b. Photograph the visor at the end of the test after the droplets have evaporated from the surface of the visor.

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The visor shall pass the requirements of 3.7.3.5. Following the rain test, the visor shall pass the requirements of 3.7.1 and 3.7.2.

4.5.5.7 Humidity. The visor shall be tested in accordance with paragraph 4.5.5.1 and the procedure outlined below. The humidity shall simulate natural environmental cycles. Ensure that the wiper on the viewing port adequately clears the surface to enable the quality of pictures required. The hot humid cycle listed in table V and shown in figure 12, shall be used for testing with a test duration of 10 days (non-hazardous items). The points at which the visor will be photographed are indicated on the temperature function shown in figure 12. There are a total of 7 photographs per 24 hour period at key transition points. This schedule of photographs will result in a total of 70 pictures over the 10 day test period. The test sequence as specified below shall be followed:

Step 1 - Prepare the visor in accordance with paragraph 4.5.5.1.
Step 2 - Place the visor in the test chamber ensuring it is positioned so the entire front surface can be photographed from the viewing port without disturbing the sample. Place a white light behind the visor so the illumination is uniform throughout the sample.
Step 3 - Stabilize the chamber to the first condition of table V.
  a. Photograph the visor after the temperature/humidity stabilization.
Step 4 - Repeat test cycle from Table V for 10 days.
  a. Photograph the visor at the hours indicated with arrows on the temperature cycle, figure 12.

The visor shall meet the requirements of 3.7.3.6. Following the humidity test, the visor shall pass the requirements of 3.7.1 and 3.7.2.

4.5.5.8 Salt fog resistance. The visors shall be tested in accordance with paragraph 4.5.5.1 and the procedure outlined below. The salt used for this test shall be sodium chloride containing (on a dry basis) not more than 0.1 percent sodium iodide and not more than 0.5 percent total impurities. A 5 ± 1 percent solution shall be prepared by dissolving 5 parts by weight of salt in 95 parts by weight of water. The solution shall be adjusted to, and maintained at, a specific gravity (figure 13) by using the measured temperature and density of the salt solution. Sodium tetraborate (borax) may be added to the salt solution as a pH stabilization agent in a ratio not to exceed 0.7g sodium tetraborate to 75 liters of salt solution. The pH of the salt solution as collected as fallout in the exposure chamber shall be maintained between 6.5 and 7.2 with the solution temperature at +95°F (+35°C). Only diluted chemically pure hydrochloric acid or chemically pure sodium hydroxide shall be used to adjust the pH. The pH measurement shall be made electrometrically or colorimetrically.

Step 1 - Prepare the visor in accordance with paragraph 4.5.5.1.
Step 2 - Adjust the test chamber temperature to 95°F (35°C)
  a. Photograph the visor.
Step 3 - After introduction of salt fog, the visor shall remain in the chamber for 48 hours.

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Step 4 - Remove the visor from the chamber.
a. Wash and photograph he visor.
Step 5 - The visor shall be stored at ambient temperature for 48 hours
a. Photograph the visor at the end of this period.

Following the salt resistance test, the visor shall pass the requirements of 3.7.1, 3.7.2 and 3.7.3.7.

4.5.5.9 Chemical resistance. The visor shall be exposed to the chemicals listed in 3.7.3.8 for a 24 hour period at ambient room temperature conditions. The chemicals shall be contained by sealing an o-ring (0.5 inch or 0.75 inch diameter) to either the concave or convex surface of the visor using a silicone grease. Several drops of the chemical shall be puddled on the surface of the visor. At the end of the test, the surface shall be thoroughly cleaned using a mild detergent and water followed by drying with a soft cloth or lens tissue. The visor shall then be visually inspected for visible damage and shall pass the requirements of 3.7.1 and 3.7.2 ensuring the same areas that were exposed to the chemicals are tested for these requirements.

4.5.6 Mechanical.

4.5.6.1 Ballistic and crack propagation resistance. Prior to performing this test, the visor shall be visually checked for defects such as cracks, chips and delamination or any damage that could occur as a result of a ballistic test. The ballistic test shall be conducted in accordance with the procedure outlined below using a .22 caliber, T37 shape fragment simulating projectile. The visor shall be rigidly mounted to a wooden support block having a surface conformal to the surface curvature of the visor as shown in Figure 14. The visor shall rest against a 1/2 inch wide strip of hook and pile material which may be used to hold the visor in place. The area to be impacted shall be normal to the line of fire. The distance from the muzzle of the gun or the last triggering device to the test sample may be less than four feet and should be established to minimize yaw. A witness plate, 11 x 14 inches, (279 x 356 mm), 0.002 inch thick (0.05mm) made of 2024-T3, 2024-T4 or 5052 aluminum alloy sheet shall be located 2 inches behind the target, curved and oriented to maximize the probability of perpendicular impact. Three valid impacts shall be made on the visor; one in the center and one in each of the critical vision areas. The ballistics test shall be considered valid only if it meets any of the following:

a. The impact velocity of the projectile is between 550 feet per second and 560 feet per second.
b. The impact velocity of the projectile is less than 550 feet per second and the impacted visor fails to meet the requirements in 3.7.4.1.
c. The impact velocity of the projectile is more than 560 feet per second and the impacted visor meets the requirements in 3.7.4.1.

The visor shall pass the requirements of 3.7.4.1.

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4.5.6.2 Abrasion resistance. Before and after the surface abrasion procedure, the transmittance and haze shall be determined as specified in 4.5.4 and 4.5.4.4. The surface abrasion procedure shall be performed as specified in ASTM D1044 or by an equivalent method using the instrument illustrated in figure 15. The standardized eraser shall be held approximately normal to the surface and rubbed across the surface of the coated visor over the same path from one point to another for 20 complete cycles with a force of 2.0 to 2.5 pounds continuously applied. Wherever possible, rubs of about 1 inch length are preferred. After the rubbing has been completed, the visor shall be thoroughly cleaned to remove dirt, film, fingermarks and grease marks using a mild detergent and distilled water followed by drying with a soft cloth or lens tissue. The visor shall pass the requirements specified in 3.7.4.2.

4.5.6.3 Adhesion of coating. Immediately upon completion of the humidity test and removal from the humidity chamber, the samples shall be dried with a cotton cloth and then subjected to the adhesion test. A test tape, with an adhesion rating of 40 ounces per inch of width, shall be firmly applied to the front surface of the visor. The area covered shall be 2 square inches. The tape shall be removed using a snapping motion applied 90 degrees to the surface. The visor shall pass the requirements specified in 3.7.4.3.

4.5.6.4 Windblast resistance. The laser protective visor assembly shall be installed on a helmet with an oxygen mask as worn. The visor shall be placed in a full down and locked position. There shall be a minimum gap between the visor and the oxygen mask. This assemblage shall be fitted to a head form securely mounted to a test platform and shall be subjected to windblast tests as follows:

For the EEU-8/P and EEU-15/P the visor assemblies shall be exposed to windblast velocities of 600 ± 60 KEAS at each of the following attitudes in the order listed:

a. head on/ head up
b. rotated 45° starboard/head up
c. head on/tilted 45° aft
d. rotated 45° port/tilted 45° aft

For the EEU-13/P and EEU-14/P the visor assemblies shall be exposed to windblast velocities of 450 ± 20 KEAS at each of the following attitudes in the order listed:

a. head on
b. rotated 45° yaw to right
c. rotated 45° yaw to left
d. 30° pitch aft

The time of exposure to this maximum range shall be 300 ± 50 ms, and the rise time to reach peak velocity shall be 125 ± 20 ms. The total windblast duration shall be at least 3.0 seconds. The helmet assembly and visor assembly shall pass the requirements of 3.7.4.4.

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4.5.7 Support equipment. The overlay template shall be the correct size and shape when compared to the visor it is designed for. It shall meet the requirements of 3.6.

4.5.8 Surface and other defects examination procedure. The visors shall be visually examined at a distance of 10 to 14 inches with fluorescent backlighting and broad diffuse backlighting. The fluorescent source shall be a single 40 watt, 2 foot tube. The fluorescent backlight shall be mounted in front of a black matte background. In addition, a diffusely scattering white surface shall be illuminated by the same fluorescent source. The visors shall be examined in transmission against the fluorescent tube for coating defects, specks, particulate matter, etc. The visor shall be viewed in transmission against the diffuse, illuminated white source for streaks, striations and non-uniformity of coloration. A suggested configuration is given in figure 16.

4.5.8.1 Surface defects. The visors shall be examined as specified in 4.5.8 using a template to define the critical area of vision, the non-critical area of vision and the area outside the area of vision as shown in figure 8. The visors shall meet the requirements of 3.10.1.

4.5.8.2 Other defects.

4.5.8.2.1 Visual examination. The visors shall be examined as specified in 4.5.8. The visors shall meet the requirements of 3.10.2.1 through 3.10.2.6.

4.5.8.2.2 Sharp edge or burr. The visors shall be physically examined for sharp edges or burrs. The examination shall consist of donning a pair of clean cotton gloves and running the palm along the edge of the visor. The detection of a sharp edge or burr shall be cause for rejection as specified in 3.10.2.7.

5.0 PACKAGING.

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contracting the responsible packaging activity.

6. NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The visors covered by this specification are intended for day/night use for eye protection against exposure to multiple discrete laser wavelengths.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation and if required the specific issue of individual documents referenced (see 2.2).

c. Instructions for requesting classified information. (The applicable wavelengths and the required densities to be supplied only upon submission of a request from the manufacturer which is accompanied by a secret clearance.)

d. Whether first article inspection is waived (4.3.1)

e. The number of first article visors required and lot size (4.3.1).

f. Name and address of first article inspection laboratory (see 4.3.1), name of the Government activity responsible for conducting the first article inspection program (see 4.3.1.1) and the conformance inspection program (see 4.4).

g. Packaging requirements (see 5.1).

6.3 First article testing. When first article testing is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first article item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.3.1. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of production qualification test results and disposition of the production qualification items. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Subject term (key word) listing.

Eye protection
Laser eye protection (LEP)
Radiation

Custodians:
Army - AV
Navy - AS

Preparing Activity
NAVY-AS
(Project No.8475-0234)

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TABLE I. First article inspection.

Characteristic	Requirement Paragraph	Inspection paragraph
Materials	3.2	4.5.1
Design and dimension	3.3	4.5.2
Support equipment	3.6	4.5.7
Optical performance		
Vertical prismatic deviation	3.7.1.1.1	4.5.3.1
Horizontal prismatic deviation	3.7.1.1.2	4.5.3.2
Spherical & cylindrical power	3.7.1.1.3	4.5.3.3
Optical distortion in critical areas	3.7.1.1.4	4.5.3.4
Photopic & scotopic transmission	3.7.2.1	4.5.4.1
Optical density & angular protection	3.7.2.2	4.5.4.2
Erythemal ultraviolet transmittance	3.7.2.3	4.5.4.3
Haze	3.7.2.4	4.5.4.4
Environmental performance		
Radiation induced damage resistance	3.7.3.1	4.5.5.2
High temperature resistance	3.7.3.2	4.5.5.3
Low temperature resistance	3.7.3.3	4.5.5.4
Temperature shock	3.7.3.4	4.5.5.5
Rain resistance	3.7.3.5	4.5.5.6
Humidity	3.7.3.6	4.5.5.7
Salt fog	3.7.3.7	4.5.5.8
Chemical resistance	3.7.3.8	4.5.5.9
Mechanical properties		
Ballistic resistance	3.7.4.1	4.5.6.1
Abrasion resistance	3.7.4.2	4.5.6.2
Adhesion of coating	3.7.4.3	4.5.6.3
Windblast resistance	3.7.4.4	4.5.6.4
Weight	3.8	
Identification of product	3.9	
Workmanship	3.10	4.5.8

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TABLE I. First Article inspection - Continued.

Characteristic	Requirement Paragraph	Inspection paragraph
Surface defects	3.10.1	
Critical viewing area	3.10.1.2	4.5.8.1
Non-critical viewing area	3.10.1.3	4.5.8.1
Area outside the area of vision	3.10.1.4	4.5.8.1
Other defects		
Cloudiness in critical area of vision	3.10.2.1	4.5.8.2.1
Crack, hole chip or break	3.10.2.2	4.5.8.2.1
Evidence of warpage	3.10.2.3	4.5.8.2.1
Striae in critical area of vision	3.10.2.4	4.5.8.2.1
Surface abrasions, dullness or roughness in the critical area of vision	3.10.2.5	4.5.8.2.1
Stain or discoloration in critical area of vision	3.10.2.6	4.5.8.2.1
Sharp edge or burr on an edge	3.10.2.7	4.5.8.2.2

TABLE II. Sampling for inspection.

Lot size	Inspection level I Sample size
1 to 32	1/
33 to 500	32
501 to 1200	74

1/ Inspect every visor.

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TABLE III. Conformance inspection.

Defect	Inspection/Test	Requirement Paragraph	Inspection Paragraph
Critical	Identification of product	3.9	Visual
	DESIGN & DIMENSION		
	Dimensional	3.4	4.5.2
	Workmanship		
	Surface defects - critical area	3.10.1	4.5.8
	Cloudiness in critical area of vision	3.10.2.1 3.10.2.2	4.5.8.2.1 4.5.8.2.1
	Crack, hole, chip or break	3.10.2.3	4.5.8.2.1
	Warping	3.10.2.4	4.5.8.2.1
	Striae or waviness in critical area of vision	3.10.2.5	4.5.8.2.1
	Surface abrasions, dullness or roughness in critical area of vision	3.10.2.6	4.5.8.2.1
	Stain or discoloration		
		3.7.2.1	4.5.4.2
	OPTICAL PERFORMANCE	3.7.2.2	4.5.4.1
	Optical density & angular protection		
	Photopic & Scotopic transmittance		
Major	WORKMANSHIP		
	Surface defects - non critical	3.10.1.3	4.5.8
	Sharp edge or burr	3.10.2.7	4.5.8.2.2
	Materials		
	OPTICAL PERFORMANCE	3.7.1.1.1	4.5.3.1
	Vertical prismatic deviation	3.7.1.1.2	4.5.3.2
	Horizontal prismatic deviation	3.7.1.1.3	4.5.3.3
	Spherical & cylindrical power	3.7.1.1.4	4.5.3.4
	Optical distortion	3.7.2.3	4.5.4.3
	Erythemal ultraviolet transmittance	3.7.2.4	4.5.5.4
	Haze		

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TABLE IV. Calculating photopic and scotopic transmittance

WAVELENGTH	$V\lambda$	$V\lambda'$	T	$V\lambda$ x T	$V\lambda'$ x T
380	.0000	.00059	.0000	.0000	.0000
385	.0001	.00111	.0000	.0000	.0000
390	.0001	.00221	.0000	.0000	.0000
395	.0002	.00453	.0000	.0000	.0000
400	.0004	.00929	.0000	.0000	.0000
405	.0006	.01850	.0000	.0000	.0000
410	.0012	.03484	.0000	.0000	.0000
415	.0022	.06040	.0000	.0000	.0000
420	.0040	.09660	.0000	.0000	.0000
425	.0073	.14360	.0000	.0000	.0000
430	.0116	.19980	.0000	.0000	.0000
435	.0168	.26250	.0000	.0000	.0000
440	.0230	.32810	.0000	.0000	.0000
445	.0298	.39310	.0000	.0000	.0000
450	.0380	.45500	.0000	.0000	.0000
455	.0480	.51290	.0000	.0000	.0000
460	.0600	.56720	.0000	.0000	.0000
465	.0739	.62050	.0000	.0000	.0000
470	.0910	.67560	.0000	.0000	.0000
475	.1126	.73370	.0000	.0000	.0000
480	.1390	.79300	.0000	.0000	.0000
485	.1693	.85090	.0001	.0000	.0001
490	.2080	.90430	.0000	.0000	.0000
495	.2586	.94910	.0000	.0000	.0000
500	.3230	.98170	.0000	.0000	.0000
505	.4073	.99840	.0000	.0000	.0000
510	.5030	.99660	.0000	.0000	.0000
515	.6082	.97500	.0000	.0000	.0000
520	.7100	.93520	.0000	.0000	.0000
525	.7932	.87960	.0000	.0000	.0000
530	.8620	.81100	.0000	.0000	.0000
535	.9149	.73320	.0004	.0004	.0003
540	.9540	.64970	.0037	.0035	.0024
545	.9803	.56440	.0221	.0217	.0125
550	.9950	.48080	.0710	.0706	.0341
555	1.000	.40150	.1437	.1437	.0577
560	.9950	.32880	.2129	.2118	.0700
565	.9786	.26390	.2608	.2552	.0688
570	.9520	.20760	.2889	.2750	.0660
575	.9154	.16020	.3014	.2759	.0483
580	.8700	.12120	.3016	.2624	.0366
585	.8163	.08990	.2915	.2380	.0262
590	.7570	.06550	.2763	.2092	.0181
595	.6949	.04690	.2596	.1804	.0122
600	.6310	.03325	.2384	.1504	.0079
605	.5668	.02312	.2118	.1200	.0049

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Wavelength	$V\lambda$	$V\lambda'$	T	$V\lambda xT$	$V\lambda' xT$
610	.5030	.01593	.1775	.0893	.0028
615	.4412	.01088	.1394	.0615	.0015
620	.3810	.00737	.1055	.0402	.0008
625	.3210	.00497	.0822	.0264	.0004
630	.2650	.00334	.0677	.0179	.0002
635	.2170	.00224	.0577	.0125	.0001
640	.1750	.00150	.0492	.0086	.0001
645	.1382	.00101	.0416	.0057	.0000
650	.1070	.00068	.0347	.0037	.0000
655	.0816	.00046	.0278	.0023	.0000
660	.0610	.00031	.0207	.0013	.0000
665	.0446	.00021	.0141	.0006	.0000
670	.0320	.00015	.0083	.0003	.0000
675	.0232	.00010	.0039	.0001	.0000
680	.0170	.00007	.0015	.0000	.0000
685	.0119	.00005	.0005	.0000	.0000
690	.0082	.00004	.0002	.0000	.0000
695	.0057	.00003	.0001	.0000	.0000
700	.0041	.00002	.0000	.0000	.0000
705	.0029	.00001	.0000	.0000	.0000
710	.0021	.00000	.0000	.0000	.0000
715	.0015	.00000	.0000	.0000	.0000
720	.0010	.00000	.0000	.0000	.0000
725	.0007	.00000	.0000	.0000	.0000
730	.0005	.00000	.0000	.0000	.0000
735	.0004	.00000	.0000	.0000	.0000
740	.0003	.00000	.0000	.0000	.0000
745	.0002	.00000	.0000	.0000	.0000
750	.0001	.00000	.0001	.0000	.0000
755	.0001	.00000	.0000	.0000	.0000
760	.0001	.00000	.0001	.0000	.0000
Totals:	21.3714	19.4142		2.6886	.4659

NOTE: The numbers in bold type shall be replaced by measurements made on the device under test.

TABLE IV. Calculating photopic and scotopic transmittance - Continued

The example is for a multiple-wavelength laser protection device.

Symbols:
- $V\lambda$ = Photopic relative luminosity (CIE 1924 standard)
- $V\lambda'$ = Scotopic relative luminosity (CIE 1925 standard)
- T = Transmittance of device under test (bold type)
- T_p = Percent photopic transmittance
- T_s = Percent scotopic transmittance

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Example: 1) *Multiply Vλ by T at each wavelength.*
2) Multiply Vλ′ by T at each wavelength.
3) Obtain separate totals for the numbers in VλxT, Vλ, Vλ′ xT and Vλ′ columns.
4) Sum of VλxT divided by sum of Vλ is the photopic transmittance. Multiply this by 100 to get percent:
Tp = 2.69/21.3 x 100 = 12.6%
5) Sum of Vλ′xT divided by Vλ′ is the scotopic transmittance. Multiply this by 100 to get percent:
Ts = .466/19.41 x 100 = 2.4%

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TABLE V. Modified hot humid cycle.

Time	Temp °F(±5°)	Temp °C(±3°C)	RH (%)
0	88	31	100
100	88	31	100
200	88	31	100
300	88	31	100
400	88	31	100
500	88	31	100
600	90	32	100
700	93	34	94
800	96	36	88
900	98	37	82
1000	100	38	79
1100	102	39	77
1200	104	40	75
1300	105	41	74
1400	105	41	74
1500	105	41	74
1600	105	41	76
1700	102	39	79
1800	99	37	82
1900	97	36	86
2000	94	34	91
2100	91	33	95
2200	90	32	96
2300	89	32	100
2400	88	31	100

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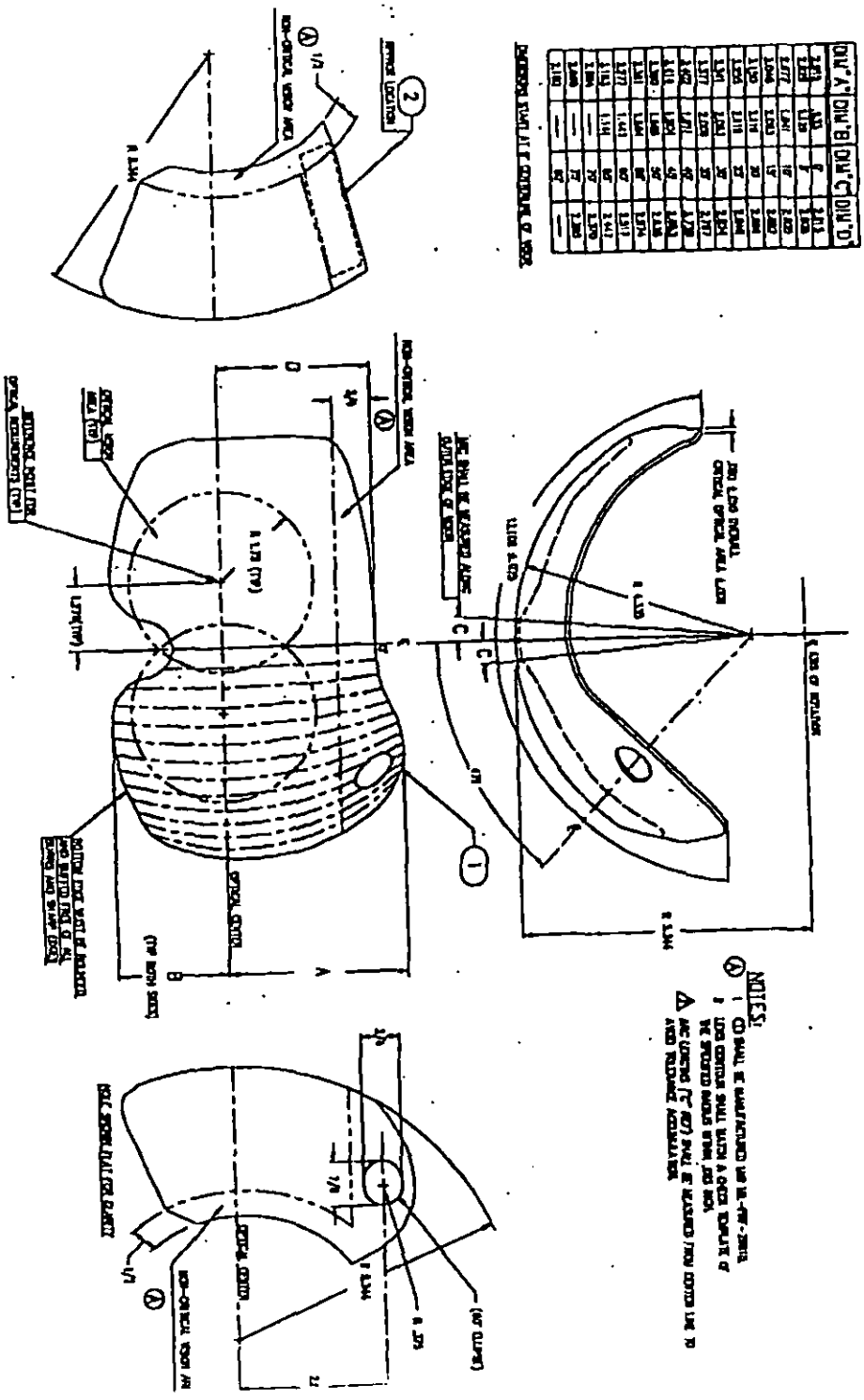


Figure 1. EEU-8/P.

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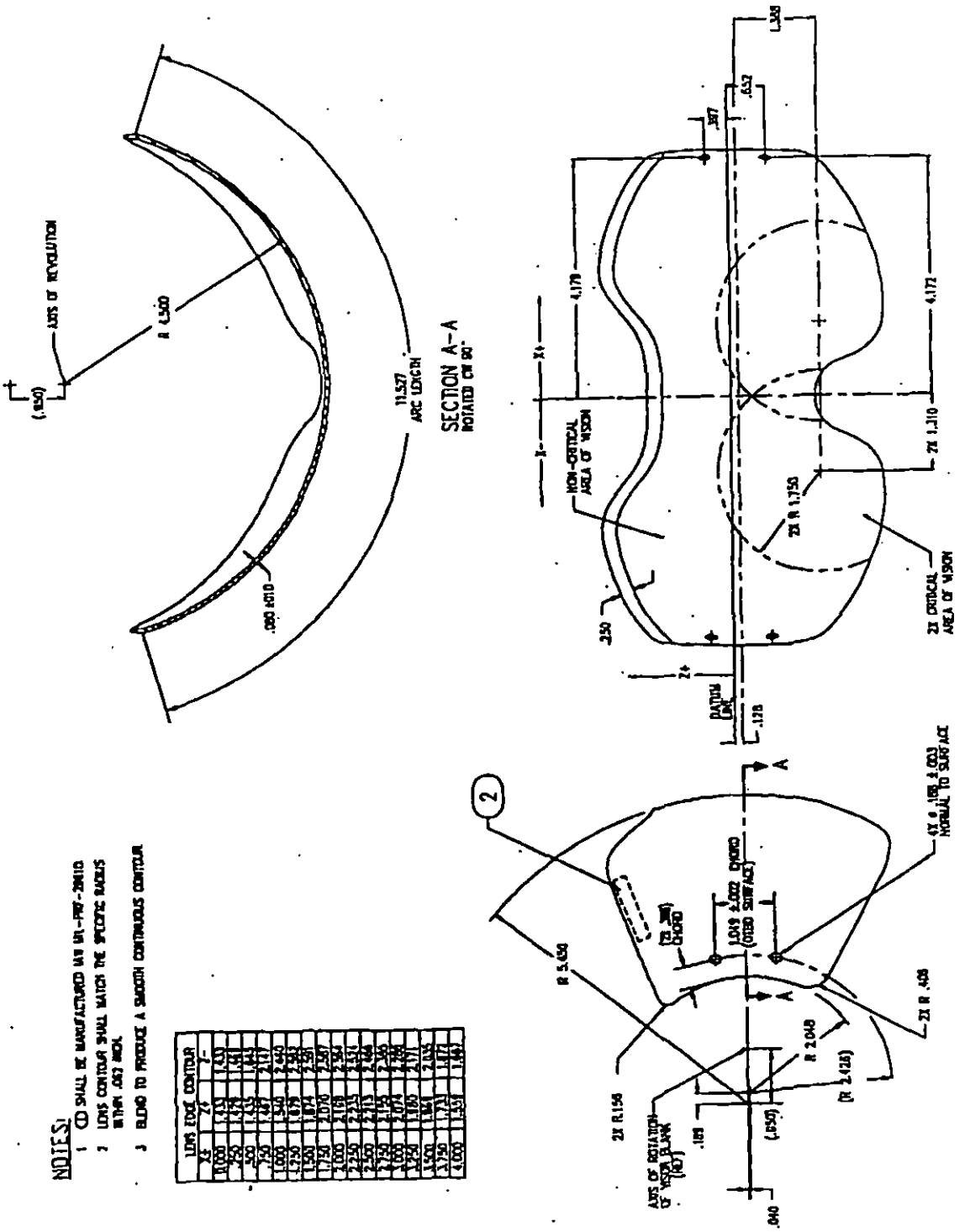


Figure 2. EEU-12/P

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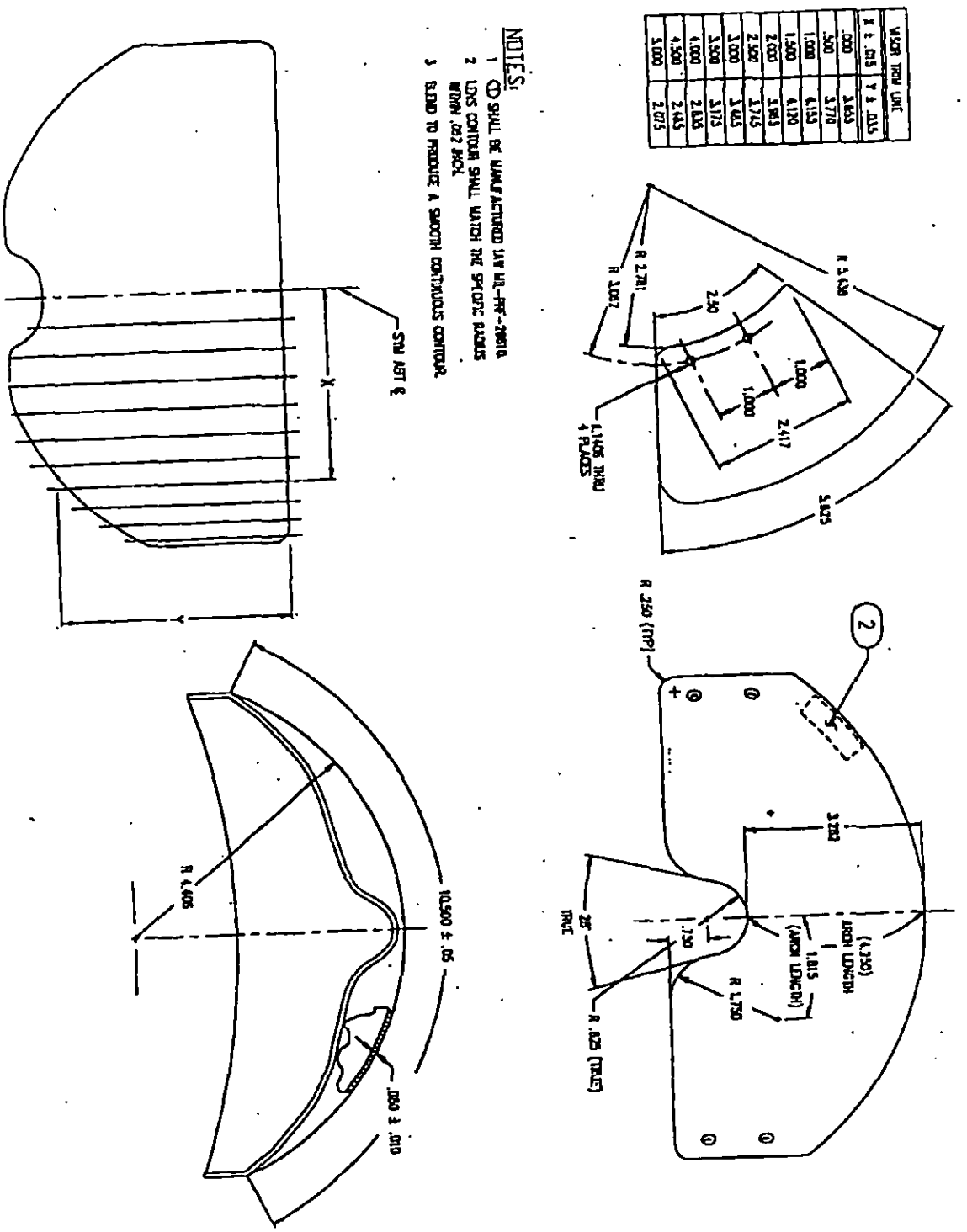


Figure 3. EEU-13/P

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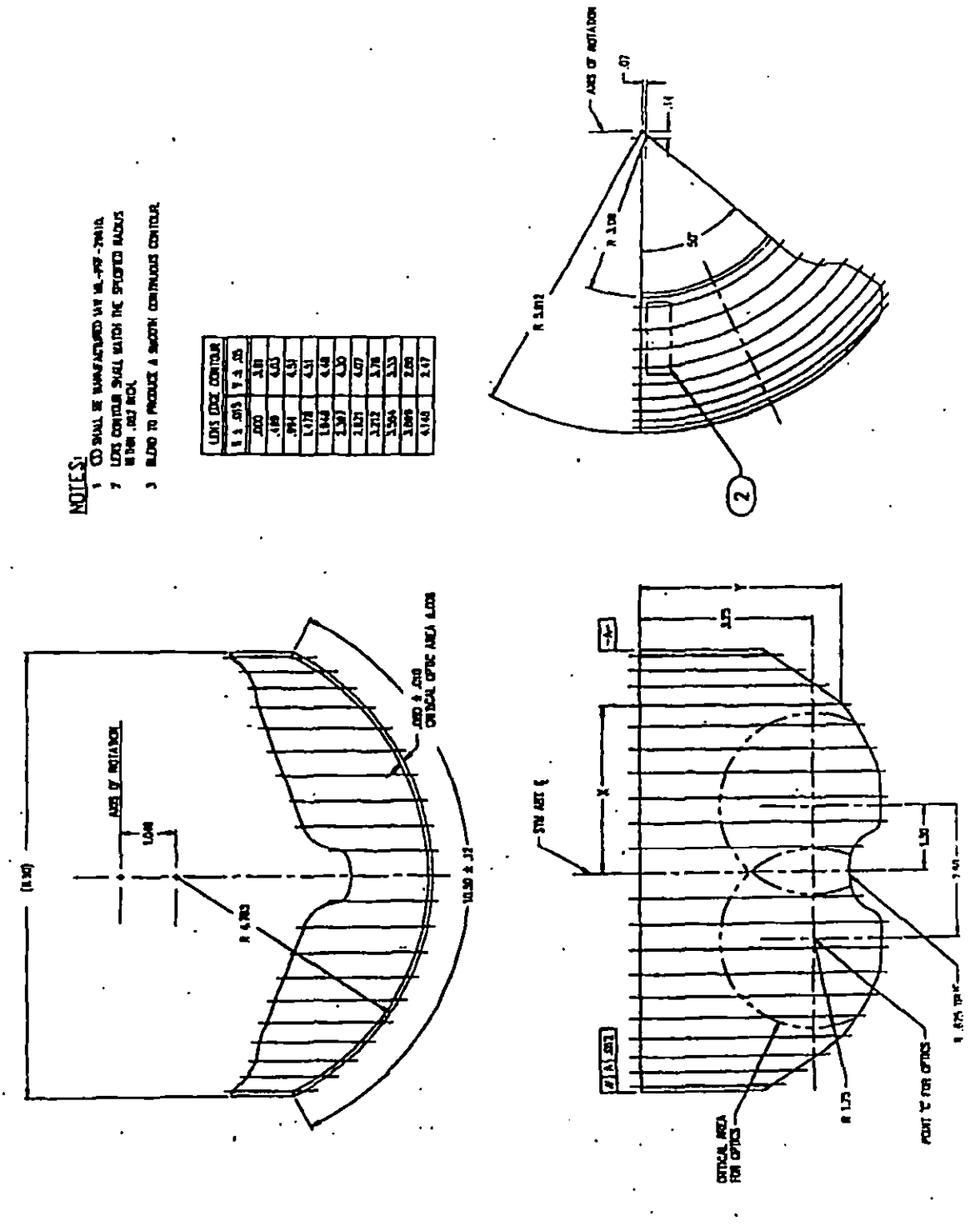


Figure 4. EEU-14/P

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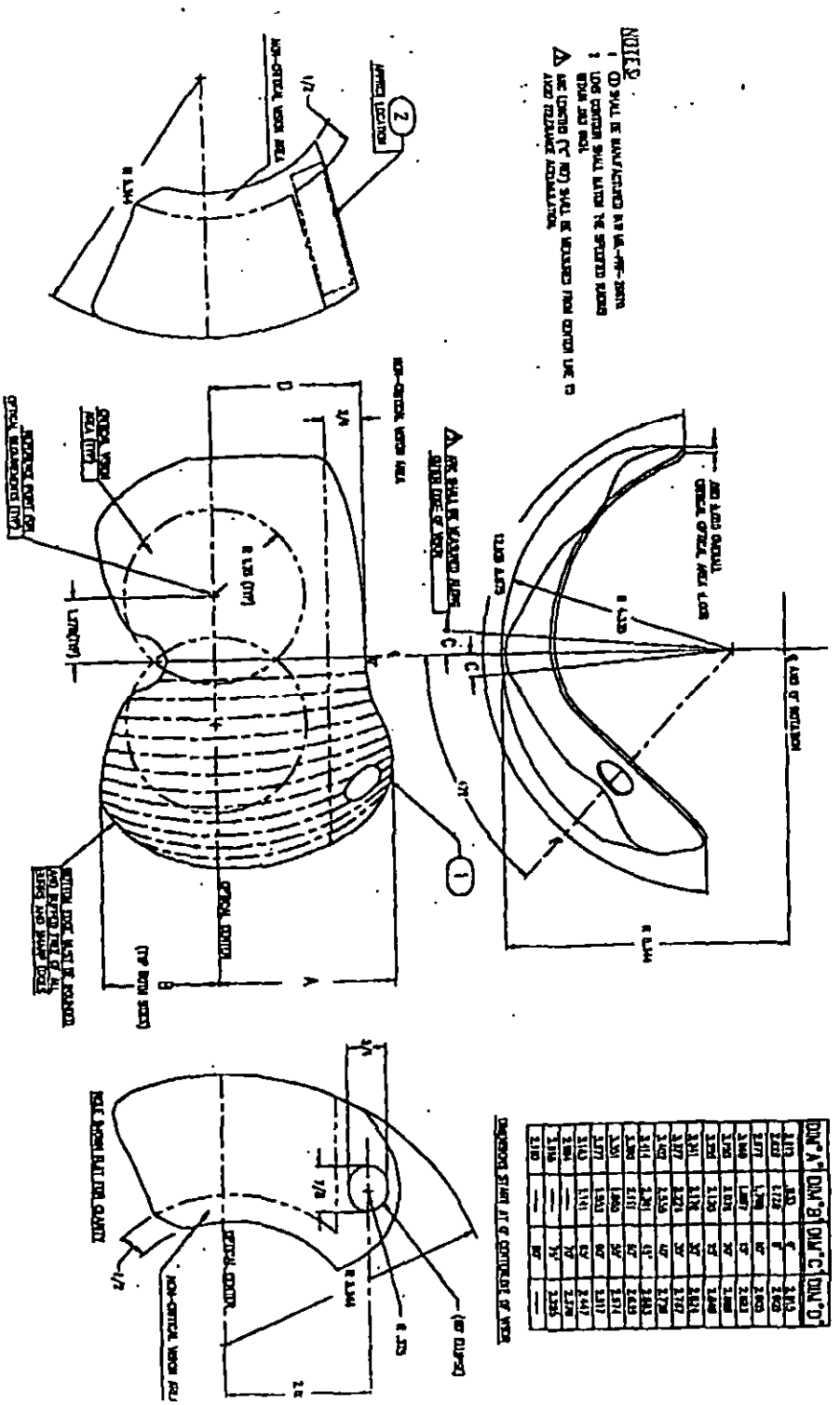


Figure 5. REU-15/P

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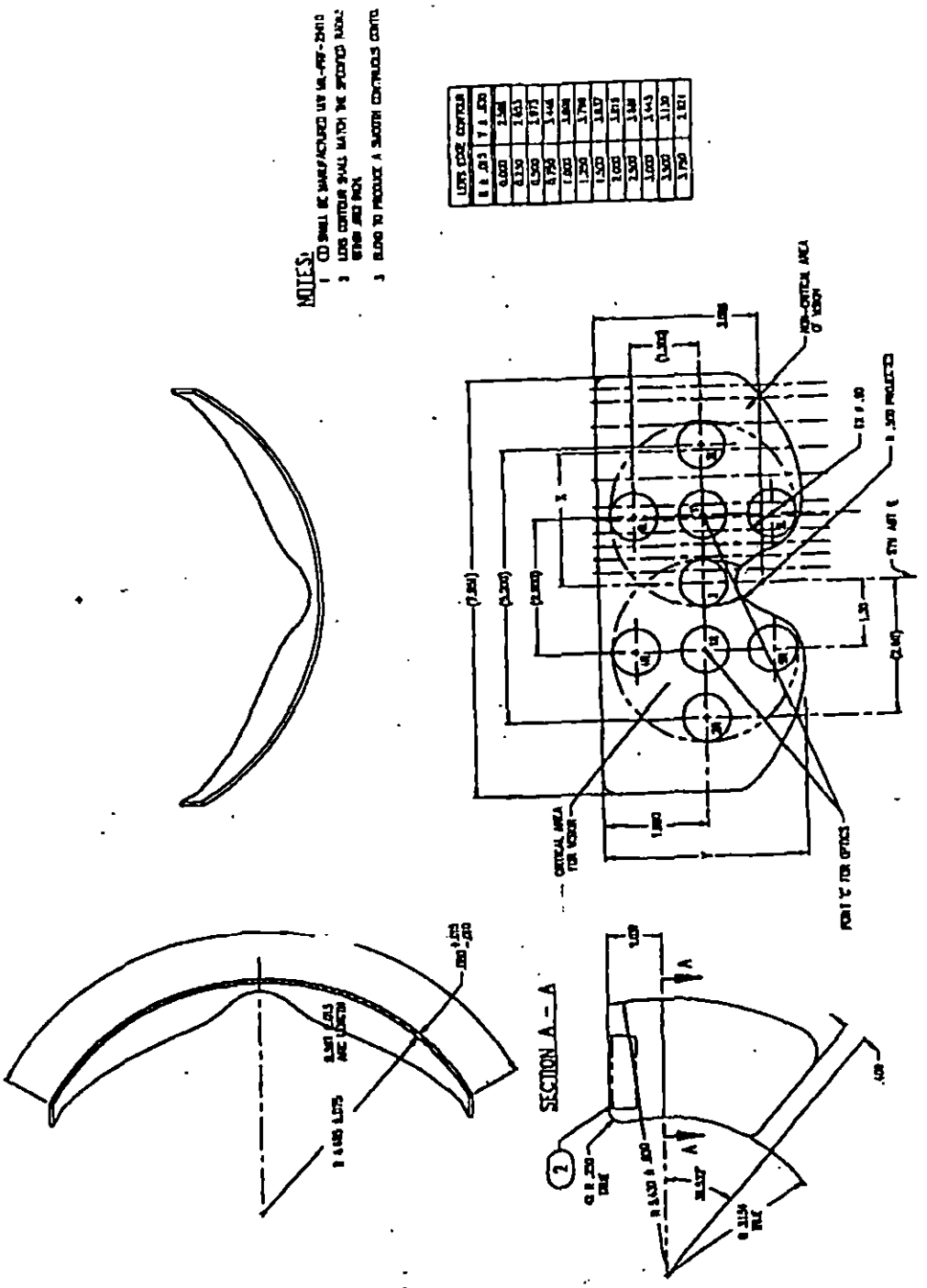


Figure 6. AHS.

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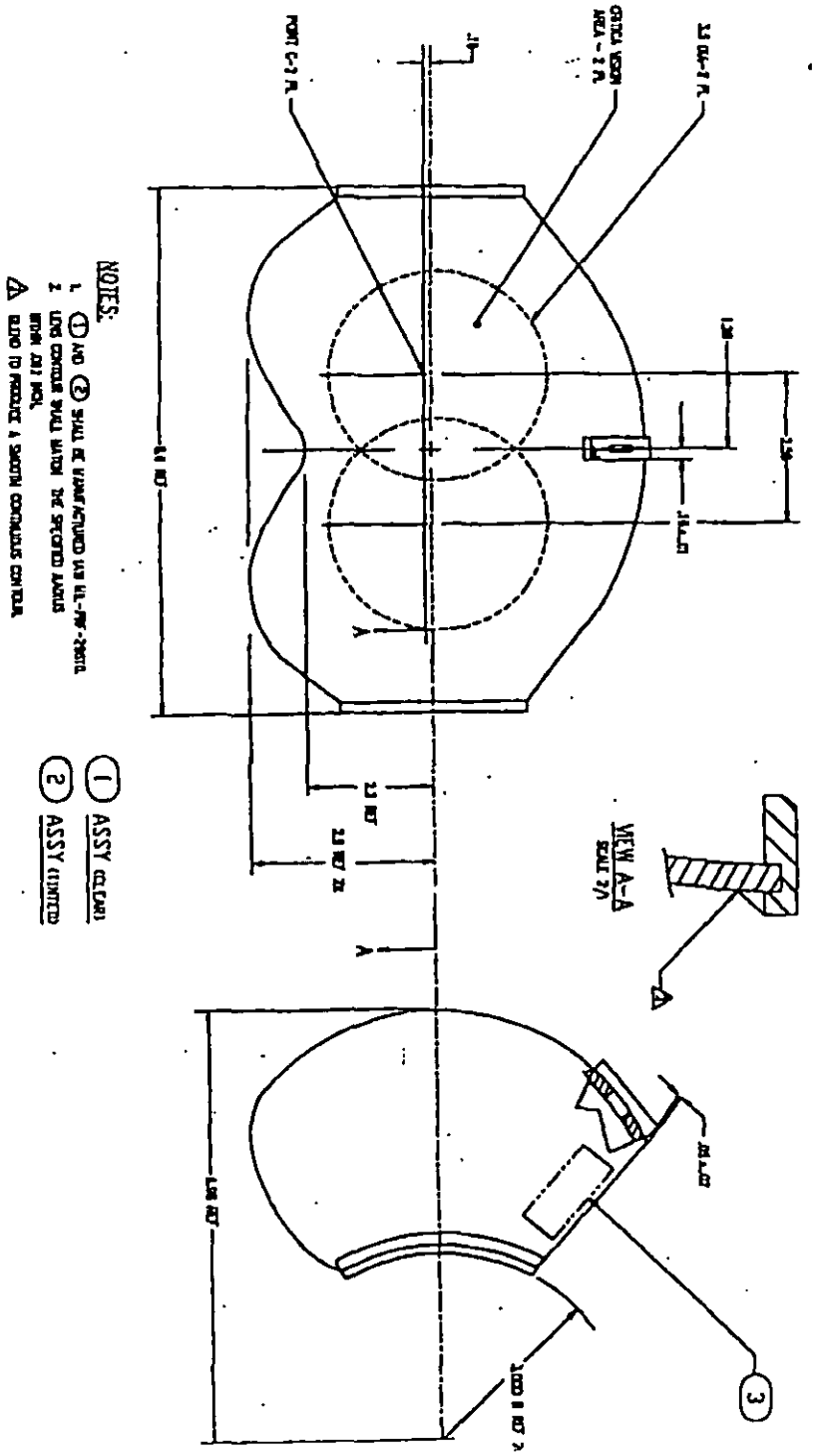
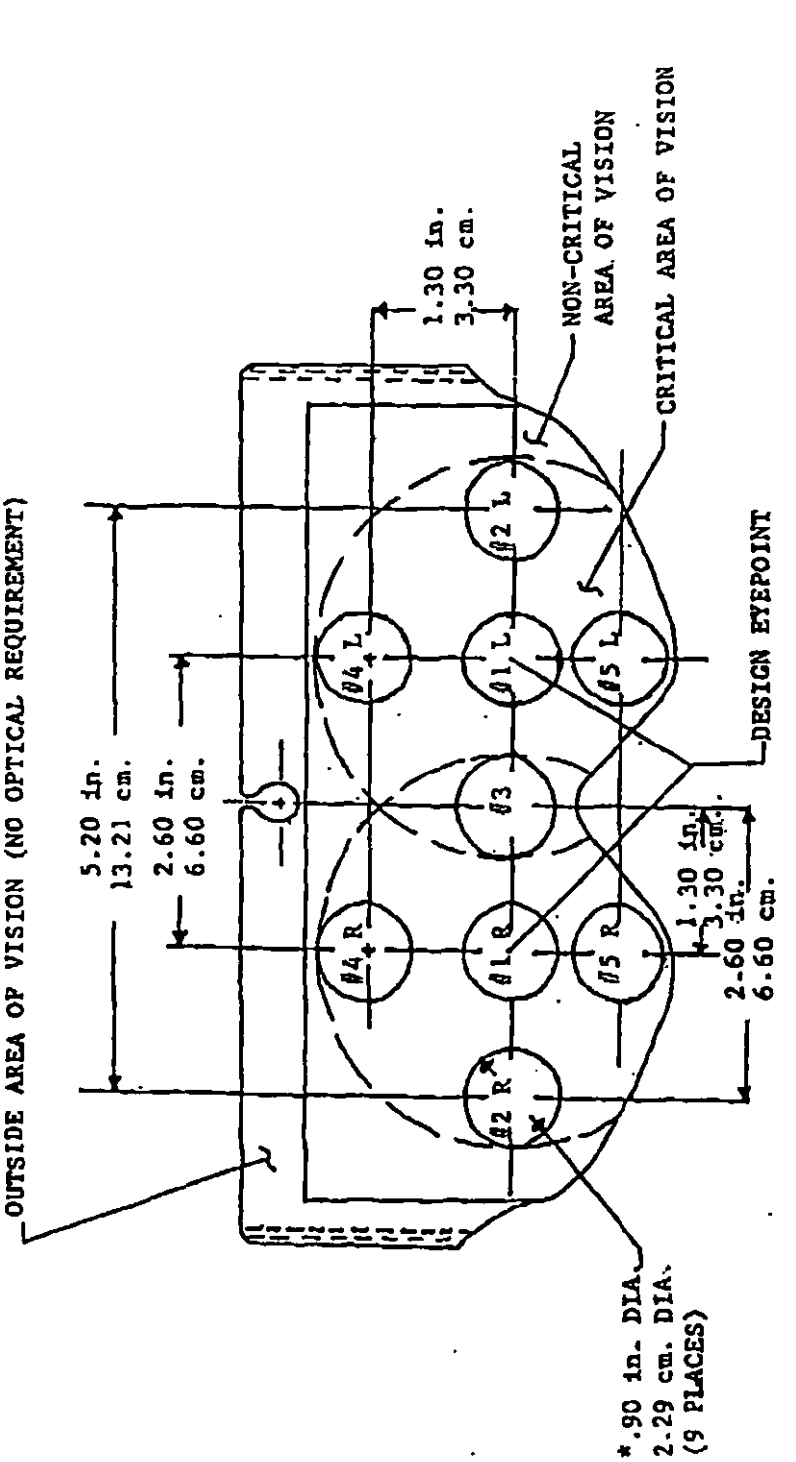


Figure 7. EOHSS

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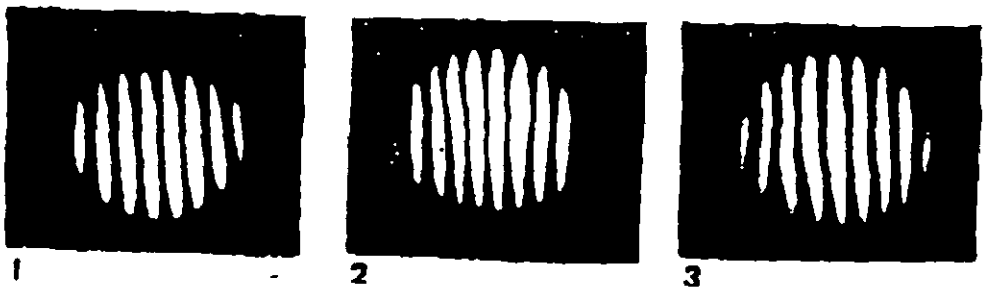


This is a generalized drawing of a typical visor for the purpose of locating test performance areas. It is not to be considered the specified configuration for this document. The numbered circles are points of choice for consistency and comparison in performing optical tests. Areas bearing the same number, for example, #5R and #5L, shall be compared with each other, except that areas #2R and #2L shall be compared with #3 when measuring refractive power or prismatic deviations.

Figure 8. Typical visor configuration

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ACCEPTABLE (Nos. 1-3) - Examples of no distortion, slight or moderate distortion



UNACCEPTABLE (Nos. 4-6) - Examples of severe distortion

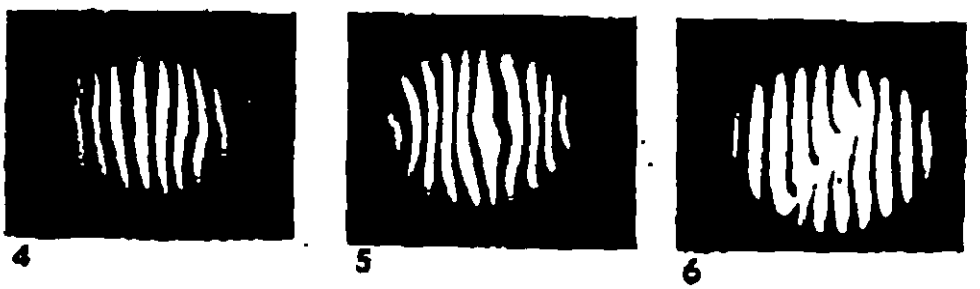


Figure 9. Distortion standards.

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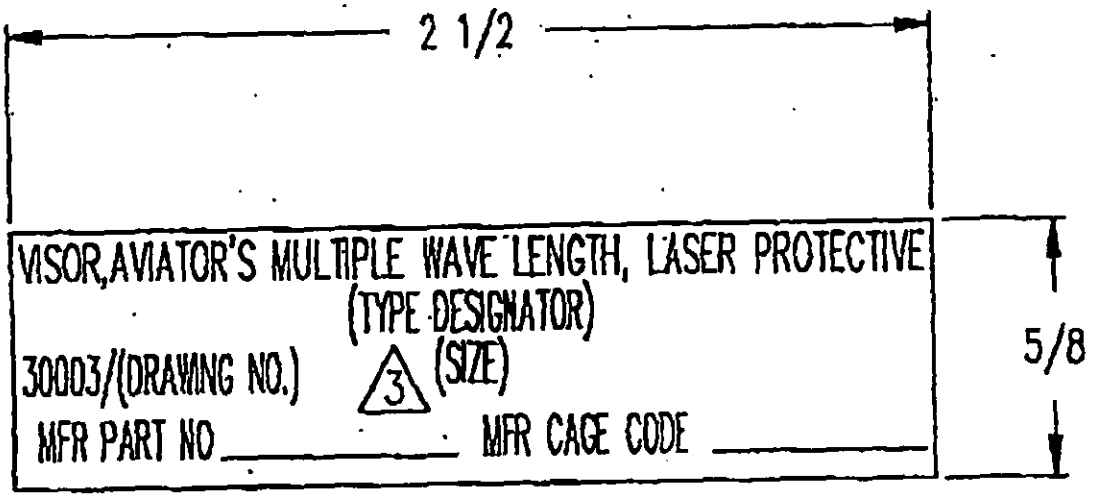


Figure 10. Product identification label

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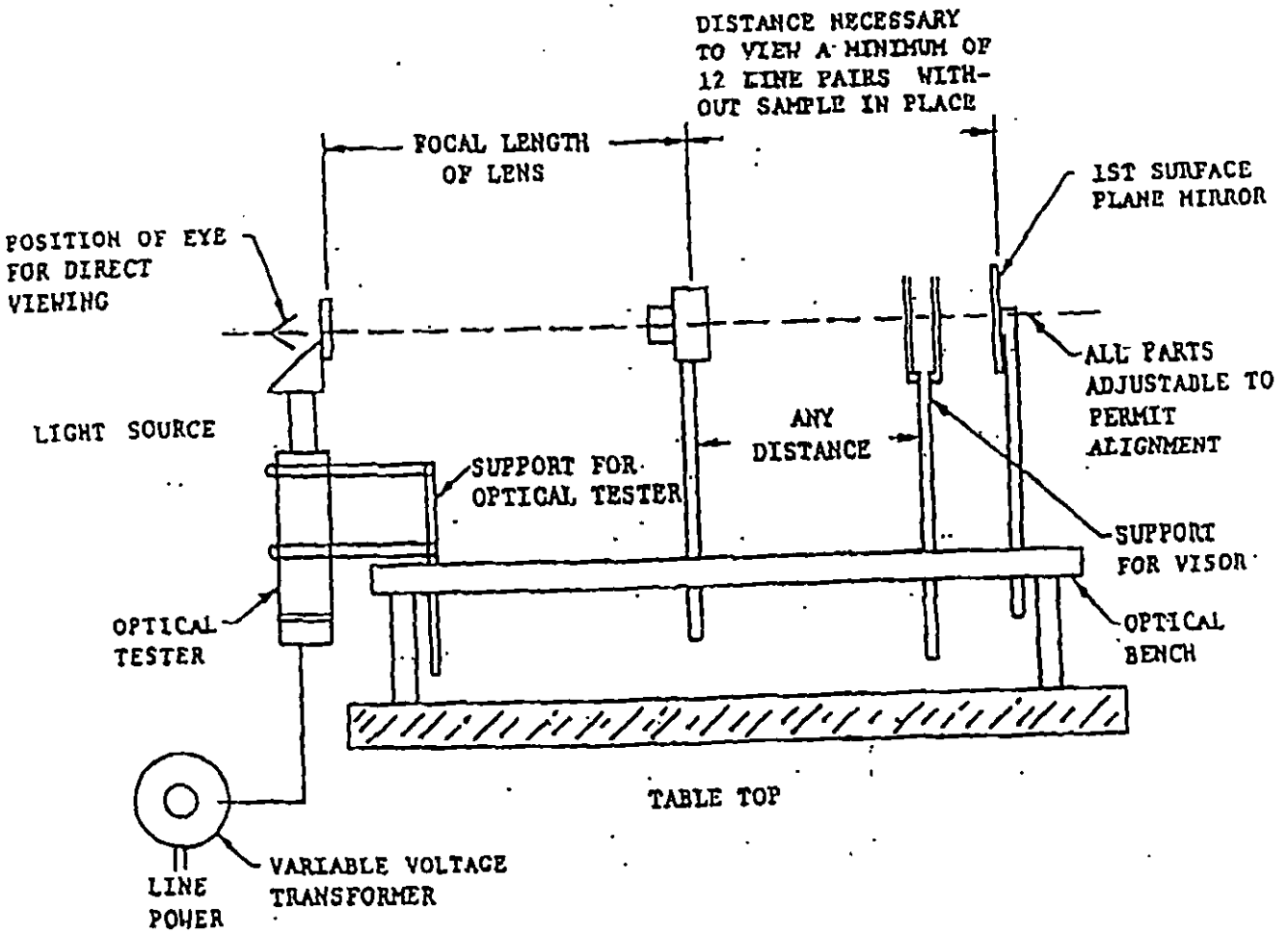


Figure 11. distortion tester

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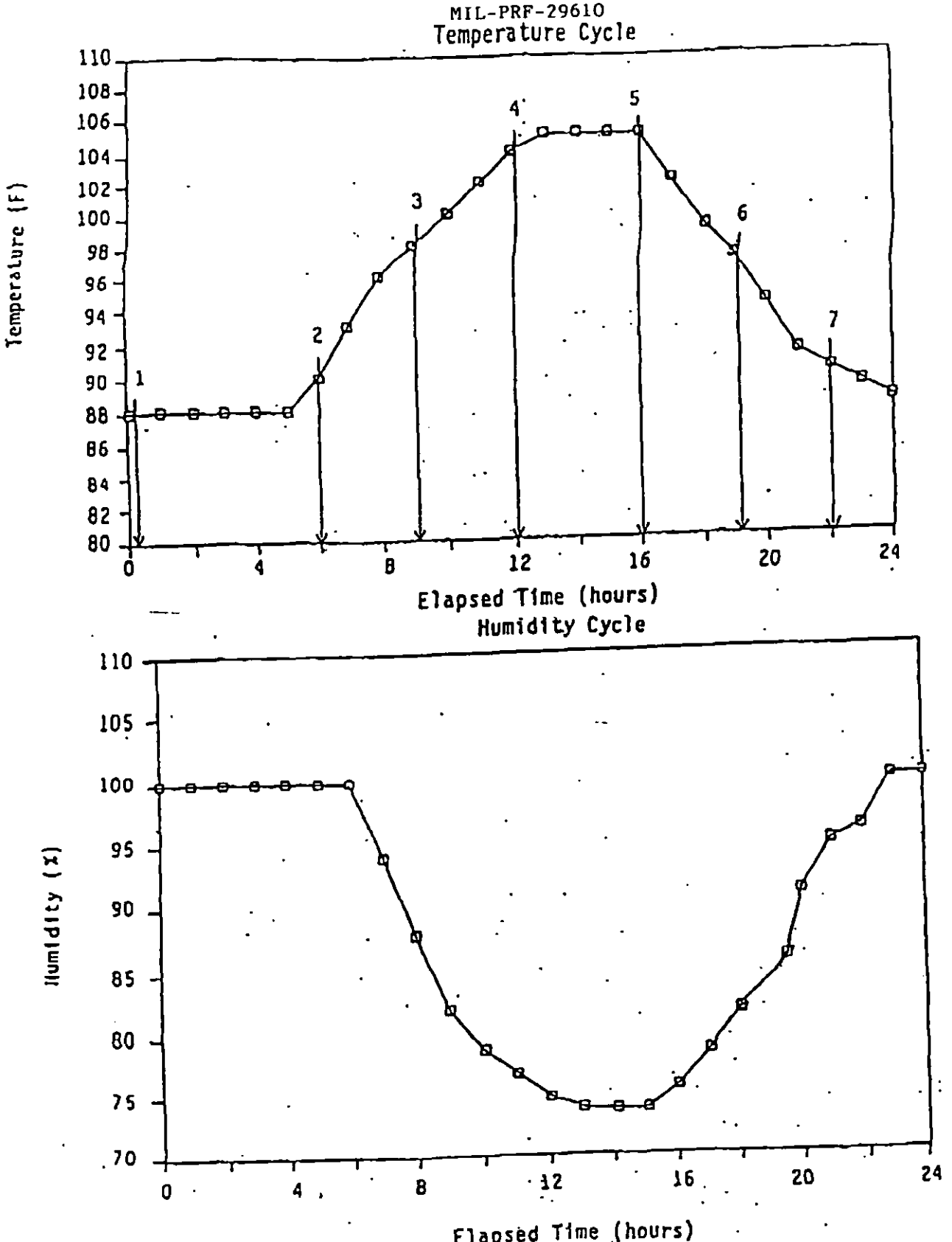


Figure 12. Hot humid cycle

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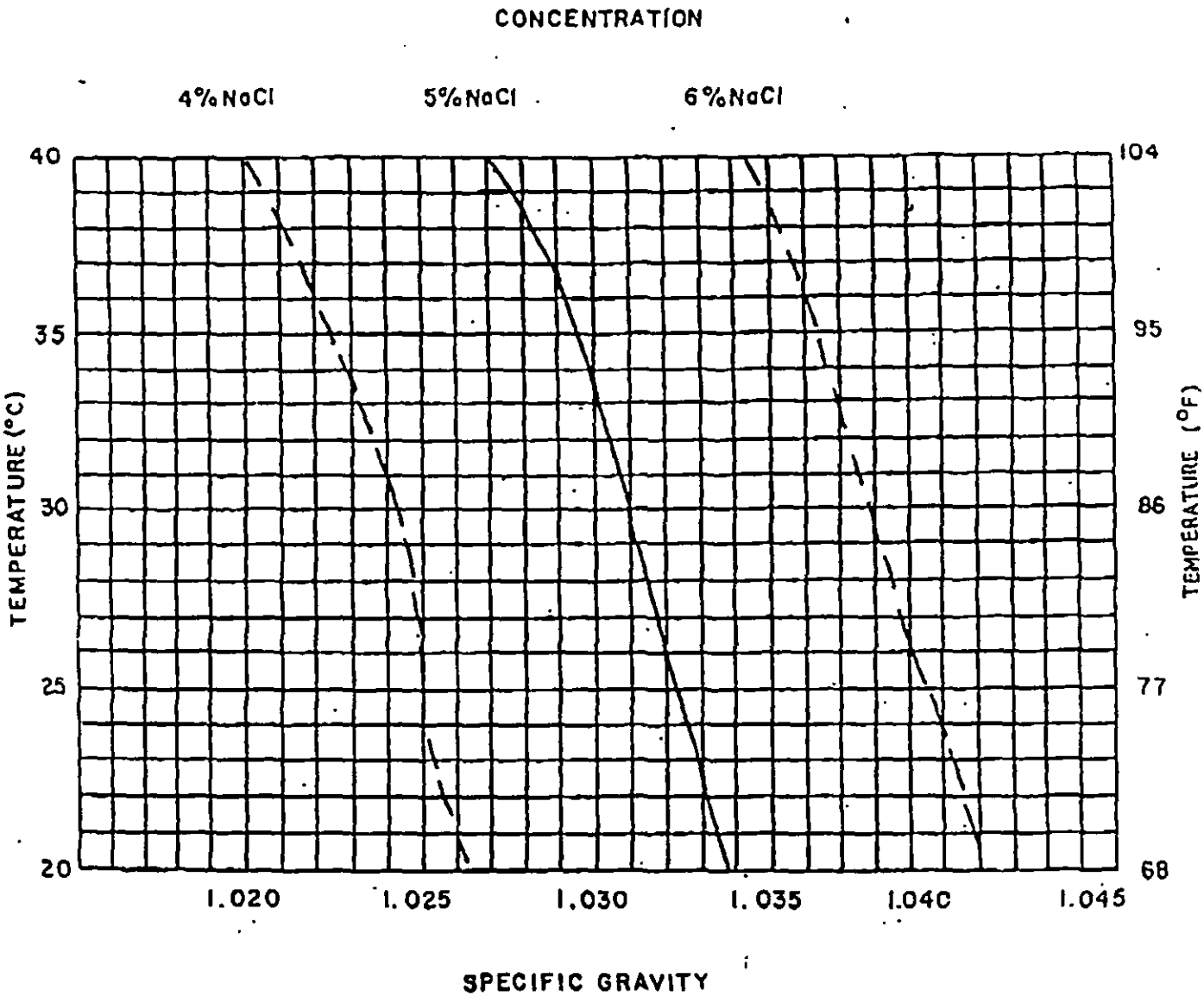
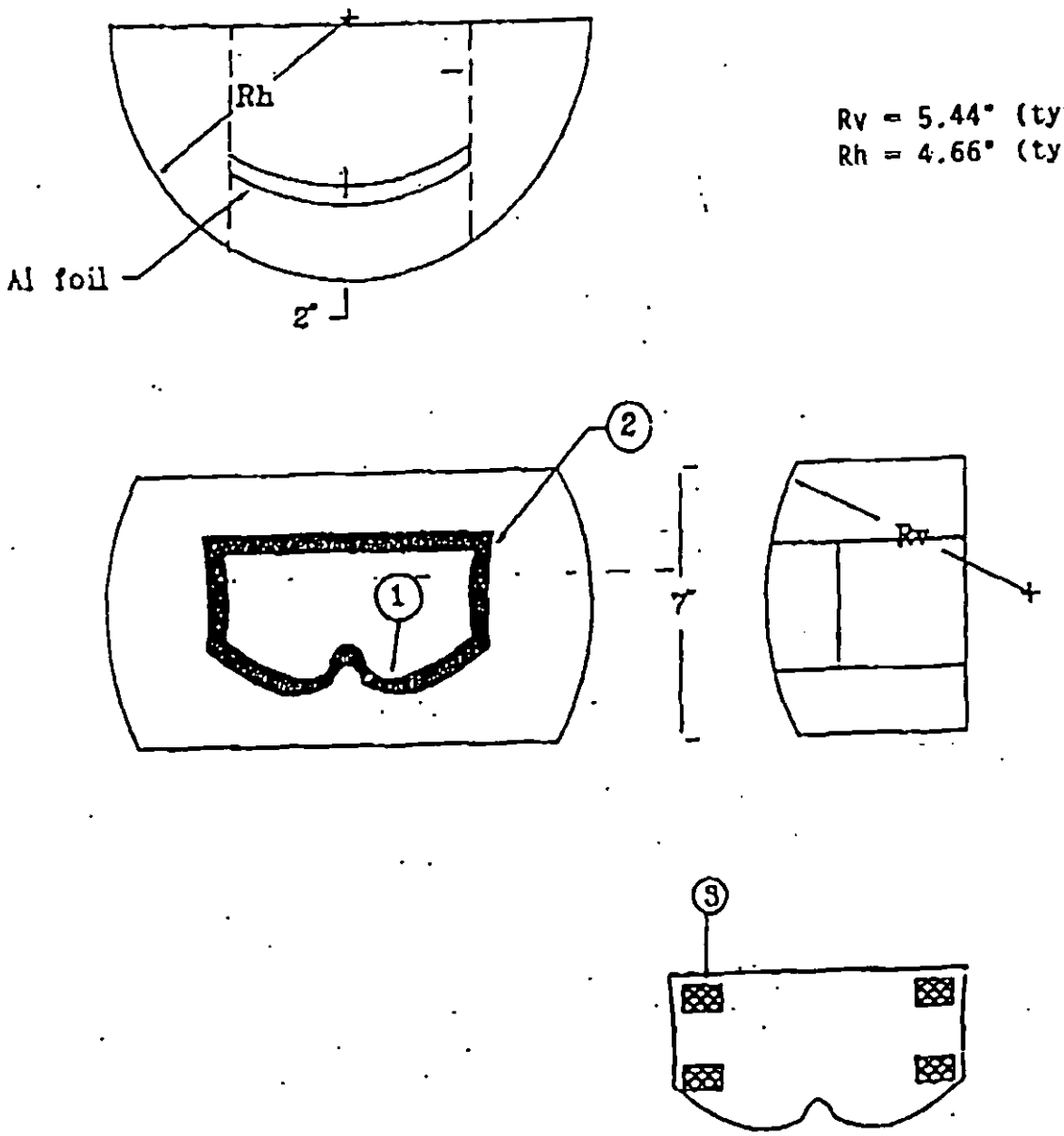


Figure 13. Variations of specific gravity of salt (NaCl) solution with temperature

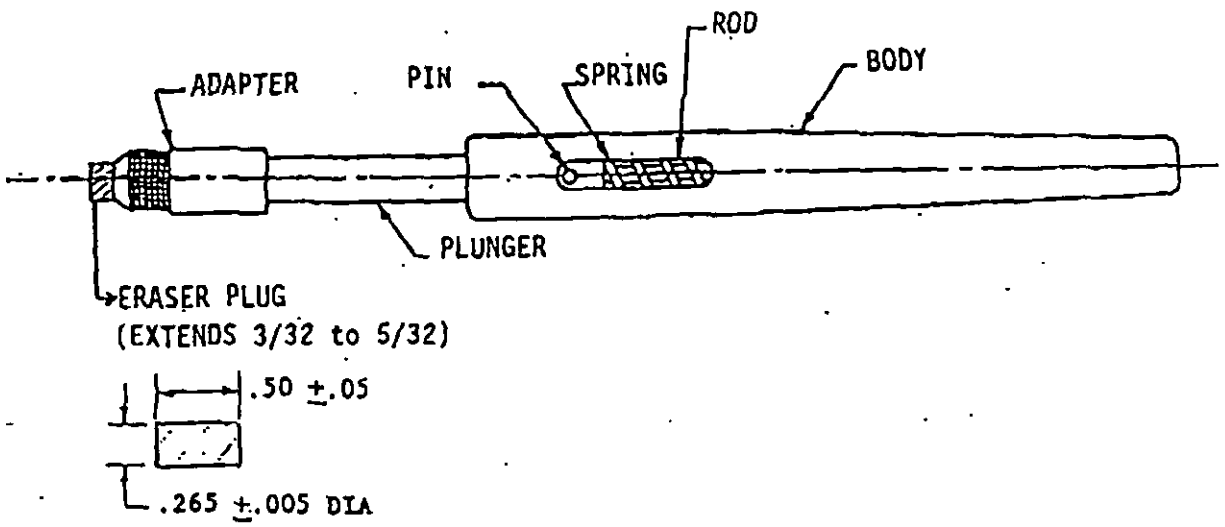
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1. Cutout 3/8" smaller than visor outline
2. Pile material
3. Hook material, self adhesive attached to visor

Figure 14. Mounting block for ballistic impact test

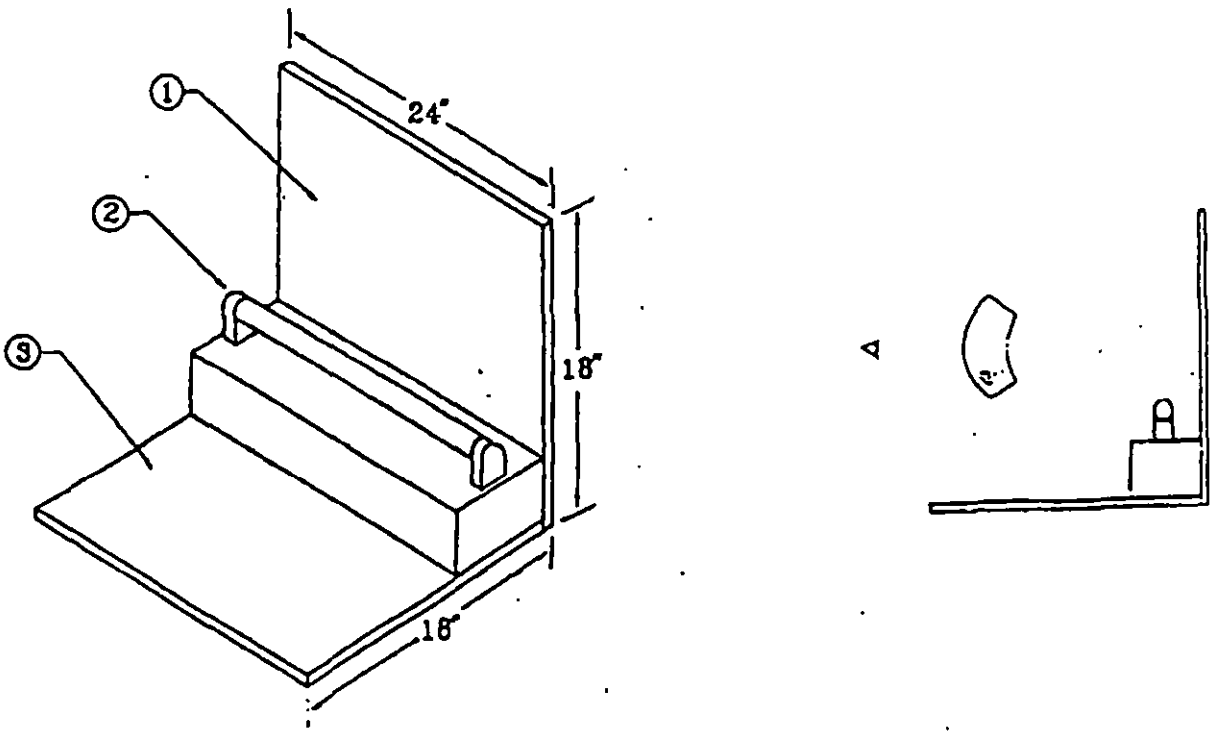
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The eraser shall be made of high-grade rubber combined with 50, plus or minus 5 percent by weight of the abrasive or filler. The abrasive shall be a fine-ground pumice. The eraser shall be a uniform mixture of abrasive and rubber, free from any excessive holes, cracks, splits, or foreign particles which might adversely affect its use. The finished eraser shall have a durometer hardness of 75, plus or minus 5.

Figure 15. Eraser abrasion tester assembly and eraser plug

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1. Black Matte Background
2. 40 Watt Fluorescent Tube
3. White Diffuse Reflector

Figure 16. Suggested inspection station

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NAVAL AIR WARFARE CENTER
AIRCRAFT DIVISION
CODE 4.1.4.2B120-3
HIGHWAY 547
LAKEHURST, NJ 08733-5100

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